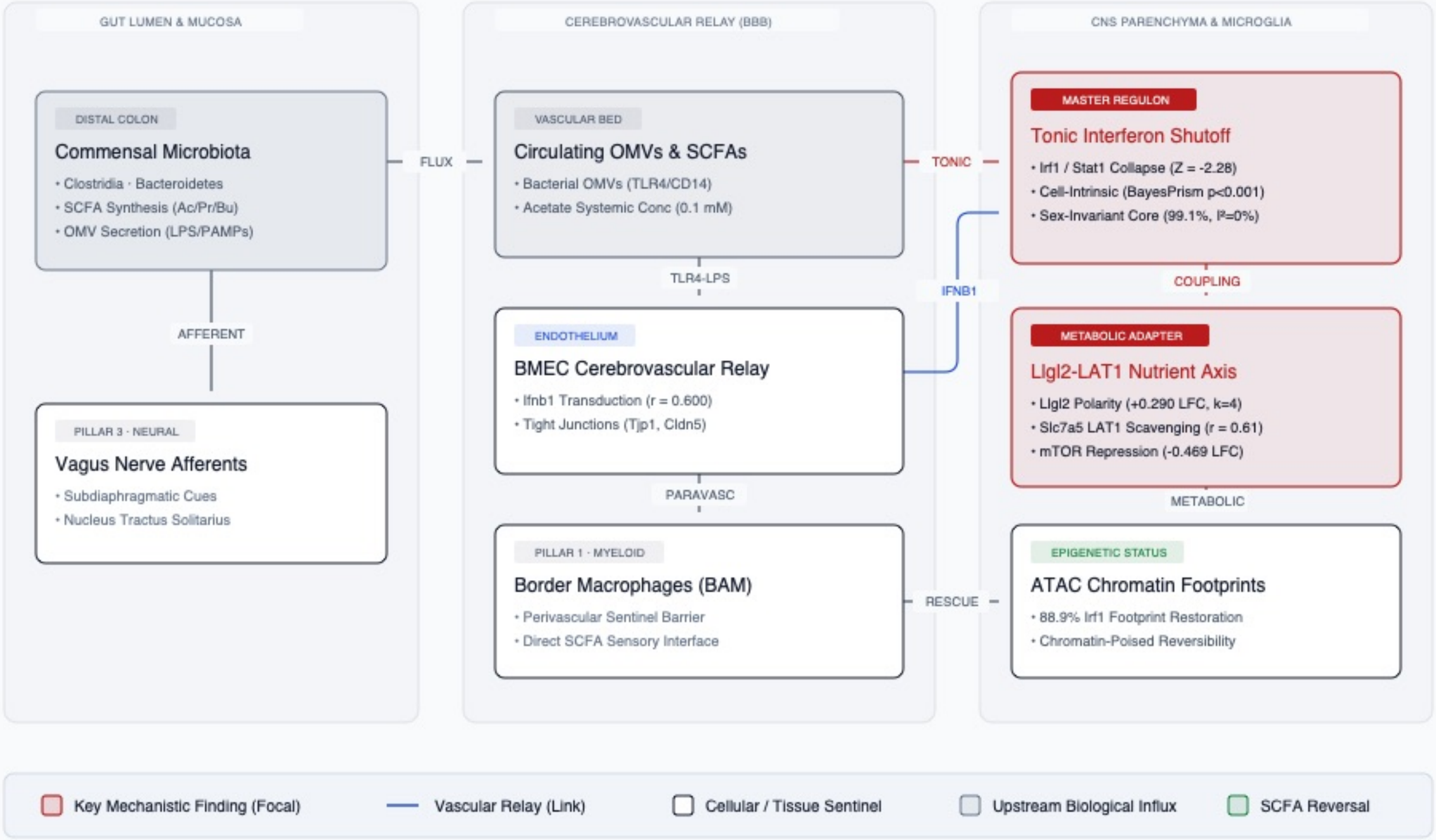

GUT LUMEN & MUCOSA
CEREBROVASCULAR RELAY (BBB)
CNS PARENCHYMA & MICROGLIA
DISTAL COLON
Commensal Microbiota
• Clostridia · Bacteroidetes
• SCFA Synthesis (Ac/Pr/Bu)
• OMV Secretion (LPS/PAMPs)
FLUX
VASCULAR BED
Circulating OMVs & SCFAs
• Bacterial OMVs (TLR4/CD14)
• Acetate Systemic Conc (0.1 mM)
TONIC
MASTER REGULON
Tonic Interferon Shutoff
• Irf1 / Stat1 Collapse (Z = -2.28)
• Cell-Intrinsic (BayesPrism p<0.001)
• Sex-Invariant Core (99.1%, I²=0%)
TLR4-LPS
IFNB1
COUPLING
AFFERENT
ENDOTHELIUM
BMEC Cerebrovascular Relay
• Ifnb1 Transduction (r = 0.600)
• Tight Junctions (Tjp1, Cldn5)
METABOLIC ADAPTER
Llgl2-LAT1 Nutrient Axis
• Llgl2 Polarity (+0.290 LFC, k=4)
• Slc7a5 LAT1 Scavenging (r = 0.61)
• mTOR Repression (-0.469 LFC)
PARAVASC
METABOLIC
PILLAR 3 · NEURAL
Vagus Nerve Afferents
• Subdiaphragmatic Cues
• Nucleus Tractus Solitarius
PILLAR 1 · MYELOID
Border Macrophages (BAM)
• Perivascular Sentinel Barrier
• Direct SCFA Sensory Interface
RESCUE
EPIGENETIC STATUS
ATAC Chromatin Footprints
• 88.9% Irf1 Footprint Restoration
• Chromatin-Poised Reversibility
Key Mechanistic Finding (Focal)
Vascular Relay (Link)
Cellular / Tissue Sentinel
Upstream Biological Influx
SCFA Reversal